

DSU08210

DATA SECURITY AND ETHICS

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Course Learning Outcomes

By the end of this course, students should be able to:

1. Critically explain and integrate core concepts of data security, privacy, and ethical data practices within data science workflows.
2. Apply the Confidentiality, Integrity, and Availability (CIA) triad to design secure and reliable data systems.
3. Identify, assess, and mitigate cyber threats, vulnerabilities, and risks affecting data infrastructures.
4. Design and implement robust security mechanisms, including encryption, authentication, access control, and intrusion detection systems.
5. Evaluate and manage cloud and distributed system security risks using industry-standard frameworks.
6. Apply ethical principles and professional codes of conduct in data collection, analysis, artificial intelligence, and decision-making processes.
7. Demonstrate responsible stewardship of personal and sensitive data, ensuring compliance with privacy laws and regulations.
8. Utilize computational tools (e.g., Python) to perform basic data security analysis and implement secure data handling techniques.
9. Analyze and resolve real-world case studies involving security breaches, ethical dilemmas, and data misuse.
10. Exhibit professional integrity, accountability, and ethical judgment in academic, research, and workplace environments.

Why Learn Data Security and Ethics?

Data Science students study Data Security and Ethics because:

- **Data is a critical asset:** Modern organizations depend on data for decision-making, innovation, and competitiveness, making its protection essential.
- **Handling sensitive information:** Data scientists work with personal, financial, health, and organizational data that must be protected from misuse or unauthorized access.
- **Rising cyber threats:** Increasing cyberattacks, data breaches, and system vulnerabilities require professionals who can secure data systems effectively.
- **Ethical responsibility in AI and analytics:** Data-driven models can introduce bias, discrimination, or harm if not ethically designed and monitored.

Why Learn Data Security and Ethics?

Data Science students study Data Security and Ethics because:

- **Legal and regulatory compliance:** Knowledge of privacy laws, data protection standards, and governance frameworks is necessary to avoid legal consequences.
- **Trust and accountability:** Ethical and secure data practices build trust among users, organizations, and society.
- **Professional competence:** Employers demand graduates who combine technical expertise with ethical awareness and security skills.
- **Real-world decision-making:** Data scientists frequently face dilemmas involving privacy, transparency, and fairness, requiring strong ethical judgment.
- **Interdisciplinary relevance:** Security and ethics intersect with cloud computing, machine learning, big data, and governance systems.
- **Future readiness:** Emerging technologies (AI, blockchain, IoT) increase both opportunities and risks, making security and ethics foundational competencies.

Lesson 01:

Data Security and Privacy

Learning Objectives

At the end of this lesson, students should be able to:

1. Define data privacy and explain its importance in digital systems.
2. Distinguish between privacy, confidentiality, and security.
3. Identify ethical issues related to data collection, storage, and sharing.
4. Explain the concept of informed consent in data usage.
5. Evaluate risks associated with unauthorized data disclosure.
6. Apply ethical principles in handling sensitive data.
7. Interpret professional codes such as ACM and IEEE ethics guidelines.
8. Analyze real-world cases involving privacy violations.
9. Propose strategies to ensure compliance with data protection policies.
10. Demonstrate responsible behavior in managing personal and organizational data.

Importance of Introducing Data Security and Ethics in this Course

Introducing this topic is essential because:

- It ensures protection of confidential and sensitive information.
- It promotes data integrity and accuracy, preventing manipulation.
- It reduces risks of cybercrime and unauthorized access.
- It addresses legal consequences of data misuse and piracy .
- It fosters ethical awareness and professional accountability.
- It prepares students to make ethical decisions under uncertainty.
- It strengthens organizational resilience through secure systems design.

Data Security

- **Data security** is basically the process of keeping certain information private
- It involves the use of various methods to make sure that data is kept confidential and safe
- **Data security** ensures the integrity and the privacy of data, as well as preventing the loss or corruption of data.

Data Integrity

- When data is processed it is usually changed in some way or another
- Data integrity describes the correctness of this change
- Safeguards are needed to make sure that the data has integrity by detecting any mistakes or malicious change to the data

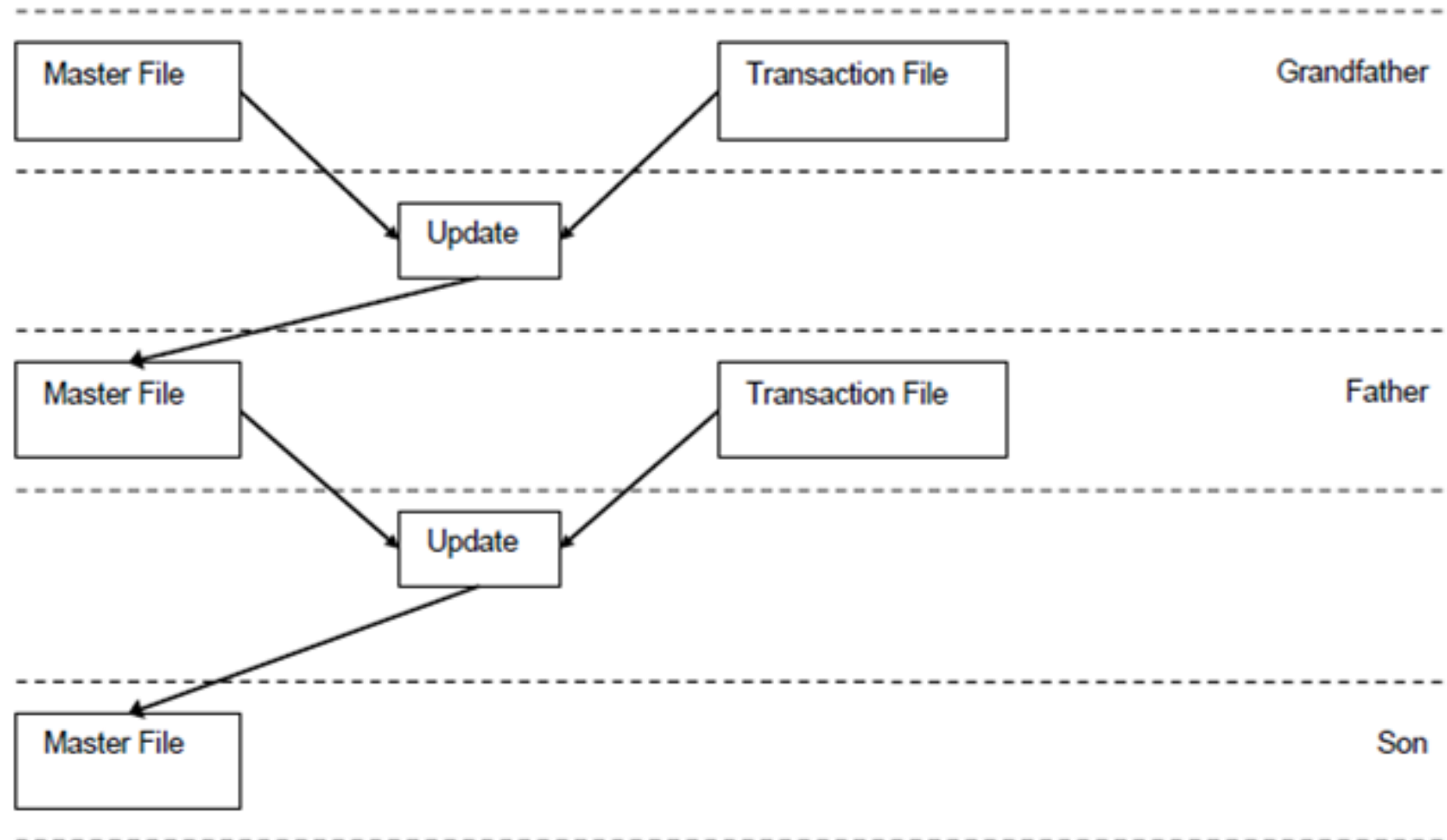
Need of Data Security

- Many businesses hold very important and confidential data
- Hence security of data is extremely important
- Data must be safeguarded all the time

Backups

- A master file stores the static data (does not change frequently) found on the database
- The transaction file keeps track of all the changes made to the database throughout the day
- At the end of the day, all the contents stored in the transaction file are transferred to the master file in order to update it
- This hierarchy is used for safety

Suppose the last Master File (Son) got corrupted by accidental deletion or corruption of data, the same Master File can be re-created by combining the father Master File with the respective Transaction File to obtain the son file once again.



Physical Security

- The most obvious choice of protecting data is to keep it in a safe locked room/building
- Protected rooms can be safeguarded by
 1. A lock-and-key
 2. ID card scanning
 3. Biometrics (retina-scan, fingerprint-scanning)
 4. Using a safe
 5. Alarm systems

Software Safeguards

- There are many software measures which can protect data.
- The following explains some of the most common approaches used now a days

IDS

- IDS stands for Intrusion Detection System
- IDS monitors the operation of the network to detect illegal operations
- The system may be
 1. server-based - detecting attacks on the operation of the file-server
 2. network based, watching the pattern of traffic across the network

User ID

- User ID stands for User Identification
- This is a unique name or code used to identify a specific user when gaining access (logging in).
- Methods of using a user ID;
 1. Passwords - words or codes known only to the user. A password is linked to a specific user ID.
 2. Personal Identification Devices - a plastic card which identifies the user and acts as an electronic key. Most cards have a magnetic stripe to store information.
 3. Personal Identification Numbers (PIN) - a number used as a password, particularly with bank cards and credit cards.

Biometric

- Biometric is when the human's features are used
- The individual's biometric is measured by a special scanner and used with the user ID
- Finding physical characteristics which cannot be copied has been difficult now a days we use
 1. fingerprints
 2. retina scans
- Face and voice recognition have not be reliable

Encryption

- Encryption makes data in a computer system illegible and makes data look meaningless
- Decryption is converting the illegible data back into its original form
- An encryption key is a code used for the encryption process
- A decryption key is needed before the data can be changed back to its original form

Digital Signature

- A digital signature makes use of encrypted data
- A digital signature is encrypted data used to show that the data being sent or read is genuine
- If the recipient of the data can correctly decrypt the digital signature then the data should be correct

Digital Certificate

- This is an encrypted message which confirms that the person is who they say they are
- A digital certificate includes a digital signature
- The certification authority, also known as a trusted service provider or a trusted third party, is a business that provides online certification facilities

Software Privacy

- Software Piracy is also very important as it stops
 1. Duplication
 2. Distribution
 3. Unauthorized use of computer software
- It is illegal to use pirated software yourself, to give it away, or worse yet to sell it

Soft – Lifting

- Soft-lifting is when a people buy software with a single license and install it on more than one PC

Hard disk loading

- Hard Disk Loading this is when computer vendors install software on a new PC without selling the software itself, this is done, to sell a fully loaded machine at very low price because the user is not charged for the software

Downloading

- Downloading software from the Internet is much quicker and easier than buying it and installing it. Many P2P (peer-to-peer) applications exist (such as Torrents) which facilitate the download of illegal software.

Software Counterfeiting

- Software Counterfeiting is when software is copied illegally and re-sold. Some counterfeited software can be very obvious because only the CD is sold, but in more “sophisticated” counterfeited software; everything will be reproduced including the box, the manuals, etc

Copyright

- Copyright is a protection that covers published and unpublished
 1. Literary,
 2. Scientific
 3. artistic works

basically whatever a person uses for expression

- The works mentioned above must be tangible or material form hence if you can see it, hear it and/or touch it; it may be protected
- Copyright laws grant only the creator the right to reproduce, prepare derivative works, distribute, perform and display the work publicly

Ethical Issues

- When you purchase software, you do not become the owner of the copyright., you are purchasing the right to use the software under certain restrictions
- Using copied or counterfeit software also means:
 1. Greater exposure to software viruses, corrupt disks, or otherwise defective software
 2. Inadequate or no documentation
 3. No warranties
 4. Lack of technical product support available to properly licensed users
 5. Ineligibility for software upgrades offered to properly licensed users.
- Software piracy is not a victimless crime, piracy denies the software developer its rightful profits and harms consumers and the industry as a whole
- All software developers, spend years creating software.

Legal Issues

- There are also serious legal issues when it comes to software privacy
- In the USA, software theft is a serious matter. If you are caught copying software, you may be held liable under both civil and criminal law
- If the copyright owner brings a civil action against you, the owner can seek to stop you from using its software immediately and can also request financial payment. The copyright owner may choose between
 1. Actual damages - which include the amount he/she has lost because of your violation
 2. Legal damages - which can be as much as \$150,000 for each program copied.
- In addition, the government can criminally prosecute you for copyright infringement, you can be fined up to \$250,000, or sentenced to jail for up to five years, or both!

Software Protection

Software developers try to protect their software by using many different protection measures.

1. **Serial Numbers:** Certain software will ask the user to input a serial number when installing the software. If the number is not inputted the software will not install
2. **Activation Keys:** After the software is installed, the user is required to enter some text (the activation key) so that the application will work. This activation key is usually obtained from the seller of the application. The user will send an e-mail with the product ID of the application, and after the seller will confirm that the software is original; he/she will send the activation key which will unlock the software.
3. **CD (or DVD) Copy Protection:** Most companies will create a special program when burning their application to the storage medium which will prevent users from copying the software
4. **Hardware Keys:** In this case a hardware device (such as a USB pen) is given with the software and for the software to be functional the USB must be connected to the machine

Software Registration

Most software is registered with the company that sells the software. The user fills in some personal details such as name, address and e-mail. This will allow the company to serve its customers better

1. **Updates:** The software company can inform its registered members with news about the product. This may include news about new program releases, new updates or new patches to the program
2. **Bonus Features:** Certain companies create bonus features to the program
3. **Discounts:** Registered members usually benefit from discounts on applications released by the same company.
4. **Technical Support:** Some companies offer technical support

Access rights

- Access rights control whether or not a particular user can use or edit a program or data file.
- Each user is assigned different rights which determine the files that can be accessed. A user may be allowed complete access to a file or may be restricted only to read the data or have no access at all.
- Network operating systems provides a way of identifying individuals (for example by a user ID and password). Each individual can only access resources the user is given privileges for by the network manager.
- Some files have additional access restrictions provided by password protection. When a user attempts to gain access to one of these files an additional password will be requested before access is allowed. This provides extra security

What is Ethics?

- Each society forms a **set of rules** that establishes the boundaries of generally accepted behavior.
- These rules are often expressed in statements about how people should behave, and they fit together to form the **moral code** by which a society lives.
- **Ethics** is the set of beliefs about right and wrong behavior.
- Ethical behavior conforms to generally accepted social norms, many of which are almost universal.
- **Virtues** are habits that incline people to do what is acceptable, and **vices** are habits of unacceptable behavior
- People's virtues and vices help define their **value system** – the complex scheme of moral values by which they live

Professional Codes of Ethics

- A **professional code of ethics** states the principles and core values that are essential to the work of a particular occupational group.
- Association of Computing Machinery ACM (founded 1947) has a code of ethics and professional conduct. See: <http://www.acm.org/constitution/code.html>
- Association of Information Technology Professionals AITP (founded 1996) – provides quality IT-related education, information on relevant IT issues, and forums for networking with experienced peers and other IT professionals. The AITP also has a code of ethics, see: <http://www.aitp.org/join/SCOH17CodeEthicsStdsCdt.pdf>

Professional Codes of Ethics

- Computer Society of the Institute of Electrical and Electronics Engineers (IEEE – CS) (founded in 1946). The Software Engineering Code of Ethics and Professional Practice:
<http://www.acm.org/serving/se/code.htm>
- Project Management Institute (PMI) – established in 1969. PMI Member Code of Ethics:
http://www.pmi.org/info/AP_MemEthStandards.pdf

Common Ethical Issues for IT Users

- **Software Piracy**: a common violation occurs when employees copy software from their work computers for use at home
- **Inappropriate Use of Computing Resources**: some employees use their work computers to surf popular Web sites that have nothing to do with their jobs.

“Half of Fortune 500 companies have dealt with at least one incident related to computer porn in the workplace over the past 12 months, according to a survey released today. Corporations are taking the problem seriously, and fired the offenders in 44% of the cases and disciplined those responsible in 41% of the instances”.

(China Martens, Survey: Computer porn remains issue at U.S. companies, Computer-world, June 21, 2005)

<http://www.computerworld.com/action/article.do?command=viewArticleBasic&articleId=102664>

Common Ethical Issues for IT Users

- **Inappropriate Sharing of Information:**
 - Organizations stored vast amount of information that can be classified as private or confidential.
 - Private data describes individual employees – for example, salary, attendance, performance rating, health record.
 - Confidential information describes a company and its operations: sales, promotion plans, research and development.
 - Sharing this information with unauthorized party, even inadvertently, has violated someone's privacy or created the potential that company information could fall into the hands of competitors.

Supporting The Ethical Practices of IT Users

- **Defining and Limiting the Appropriate Use of IT Resources**
 - Companies must develop, communicate and enforce written guidelines that encourage employees to respect corporate IT resources and use them to enhance their job performance.
 - Effective guidelines allow some level of personal use while prohibiting employees from visiting objectionable Web sites or using company e-mail to send offensive or harassing messages.

See: Widener University ITS Procedures:

http://www.widener.edu/Administration/Information_Technology_Services/Information/Procedures/6662/

Supporting The Ethical Practices of IT Users

- **Establishing Guidelines for Use of Company Software**
 - Company IT managers must provide clear rules that govern the use of home computers and associated software.
 - The goal should be to ensure that employees have legal copies of all software
- **Structuring Information Systems to Protect Data and Information**
 - Implement system and procedures that limit data access to employees who need it.
 - Employees should be prohibited from accessing the data about research and development results, product formulae, and staffing projections if they don't need it to do their job

Supporting The Ethical Practices of IT Users

- **Installing and Maintaining a Corporate Firewall**

- Firewall is a software or hardware device that serves as a barrier between a company and the outside world and limits access to the company's network based on the Internet usage policy.
- Firewall can be configured to serve as an effective deterrent unauthorized Web surfing by blocking access to specific, objectionable Web sites.
- Firewall can serve as an effective barrier to incoming e-mail from certain Web sites, companies or users
- Can be programmed to block e-mail with certain kinds of attachments, which reduces the risk of harmful computer viruses

Computer and Internet Crime

IT Security Incidents

- The security of IT used in business is very important
- Although, the necessity of security is obvious, it often must be balanced against other business needs and issues
- IT professionals and IT users all face a number of ethical decisions regarding IT security:

Ethical Decisions Regarding IT Security

- Business managers, IP professionals, and IT users all face a number of ethical decisions regarding IT security:
 - If their firm is a victim of a computer crime, should they pursue prosecution of the criminals at all costs, should they maintain a low profile to avoid the negative publicity, must they inform their affected customers, or should they take some other actions?
 - How much effort and money should be spent to safeguard against computer crime (how safe is safe enough?)

Ethical Decisions Regarding IT Security

- If their firm produces software with defects that allow hackers to attack customer data and computers, what actions should they take?
- What tactics should management ask employees to use to gather competitive intelligence without doing anything illegal?
- What should be done if recommended computer security safeguards make life more difficult for customers and employees, resulting in lost sales and increasing costs?

Challenges

- Increasing complexity increases vulnerability:
 - The computing environment has become very complex
 - Networks, computers, OS, applications, Web sites, switches, routers and gateways are interconnected and driven by hundreds of millions of lines of code
 - The number of possible entry points to a network expands continually as more devices are added, increasing the possibility of security breaches

Challenges

- Higher computer user expectations:
 - Time means money
 - Help desks are under intense pressure to provide fast responses to user's questions.
 - Sometimes forgets to verify user's identities, or to check authorization to perform a requested action

Challenges

- Expanding and changing systems introduce new risks:
 - Businesses had moved from an era of stand-alone computers to a network era – personal computers connect to networks with millions of other computers all capable of sharing information.
 - E-commerce, mobile computing, collaborative work groups, global business
 - It is increasingly difficult to keep up with the pace of technological change, successfully perform an ongoing assessment of new security risks, and implement approaches for dealing with them

Challenges

- Increases reliance on commercial software with known vulnerabilities:
 - **Exploit** is an attack on an information system that takes advantage of a particular system vulnerability. Often, this attack is due to poor system design or implementation.
 - Once a vulnerability is discovered, software developers create and issue a “fix” or **patch** to eliminate the problem. Users are responsible for obtaining and installing the patch. Any delay in installing a patch exposes the user to a security breach.
 - A rate of discovering software vulnerabilities exceeds 10 per day, creating a serious work overload for developers who are responsible for security fixes.

Challenges: Increases reliance on commercial software with known vulnerabilities:

- **A zero-day** attack take place BEFORE the security community or a software developer knows about a vulnerability or has been able to repair it.

<http://www.computerworld.com/securitytopics/security/hacking/story/0,10801,90447,00.html?f=x583>

- Malicious hackers are getting better and faster at exploiting flaws.
- The SQL Slammer worm appeared in January 2004, eight month after the vulnerability it targeted was disclosed:

<http://www.computerworld.com/softwaretopics/software/groupware/story/0,10801,89637,00.html>

Review Questions

Q1: What is data security and why is it important?

Answer: Data security is the process of protecting data from unauthorized access, corruption, or loss. It ensures confidentiality, integrity, and availability of data, which is critical for organizational operations and trust.

Q2: Explain data integrity.

Answer: Data integrity refers to the accuracy and consistency of data during processing and storage. Safeguards detect errors or malicious changes to ensure correctness .

Q3: What are common physical security measures?

Answer: Locks, ID cards, biometrics (fingerprints, retina scans), safes, and alarm systems protect physical access to data.

Review Questions

Q4: Describe encryption and its purpose.

Answer: Encryption converts data into unreadable form to prevent unauthorized access. Decryption restores it using a key.

Q5: What is software piracy?

Answer: Illegal copying, distribution, or use of software without permission, which violates copyright laws.

Q6: Explain digital signatures.

Answer: Encrypted data used to verify authenticity and integrity of transmitted information.

Q7: What are access rights?

Answer: Permissions assigned to users determining what data or systems they can access or modify.

Review Questions

Q8: What is ethics in computing?

Answer: Ethics refers to moral principles guiding acceptable behavior in computing and data use.

Q9: Identify common ethical issues in IT.

Answer: Software piracy, misuse of resources, and unauthorized sharing of confidential information.

Q10: What are the consequences of violating software laws?

Answer: Legal penalties include fines, imprisonment, and financial damages up to significant amounts

Lesson Summary

- This lesson explored the principles of **data security and ethics**, emphasizing the protection of data through physical and software safeguards such as encryption, authentication, and access control.
- It highlighted the importance of maintaining data integrity and preventing unauthorized access.
- Ethical considerations, including software piracy, misuse of resources, and privacy violations, were examined alongside legal implications.
- Professional codes of ethics guide responsible conduct, ensuring that data professionals act with integrity and accountability.

Takeaways / Lessons Learnt

- Data is a valuable and vulnerable asset requiring protection.
- Security involves both technical and human factors.
- Ethical behavior is essential in data handling and decision-making.
- Violations of data security and ethics have serious legal and societal consequences.
- Professionals must follow codes of conduct and best practices.
- Continuous learning is necessary due to evolving threats and technologies.

Acknowledgement

- This material acknowledges the contributions of scholars and practitioners in the fields of **information security and computing ethics**. Special recognition is given to professional organizations such as:
 - Association for Computing Machinery (ACM)
 - Institute of Electrical and Electronics Engineers (IEEE)
 - Association of Information Technology Professionals (AITP)
- Their ethical frameworks guide responsible computing practices.
- Appreciation is also extended to authors and publishers of foundational texts in data security and network protection, whose work supports academic and professional development.
- Additionally, the lecture content integrates real-world practices and challenges faced by organizations in safeguarding data and ensuring ethical compliance.

Reading Assignments

Students are encouraged to:

1. Study data encryption techniques and real-world applications.
2. Review case studies on data breaches and ethical failures.
3. Explore ACM and IEEE Codes of Ethics.
4. Analyze laws related to copyright and intellectual property.
5. Investigate modern cybersecurity threats and defenses.

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